Machine Learning: Decision Tree and Random Forest

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Recall for Naïve Bayes ML algorithm

Task to solve:

Likelihood Table		Buy		
		Yes	No	
Day	Weekday	4/10	1/5	5/15
	Weekend	2/10	2/5	4/15
	Holiday	4/10	2/5	6/15
		10/15	5/15	

Likelihood Table		Buy		
		Yes	No	
Free Delivery	Yes	9/10	2/5	11/15
	No	1/10	3/5	4/15
		10/15	5/15	

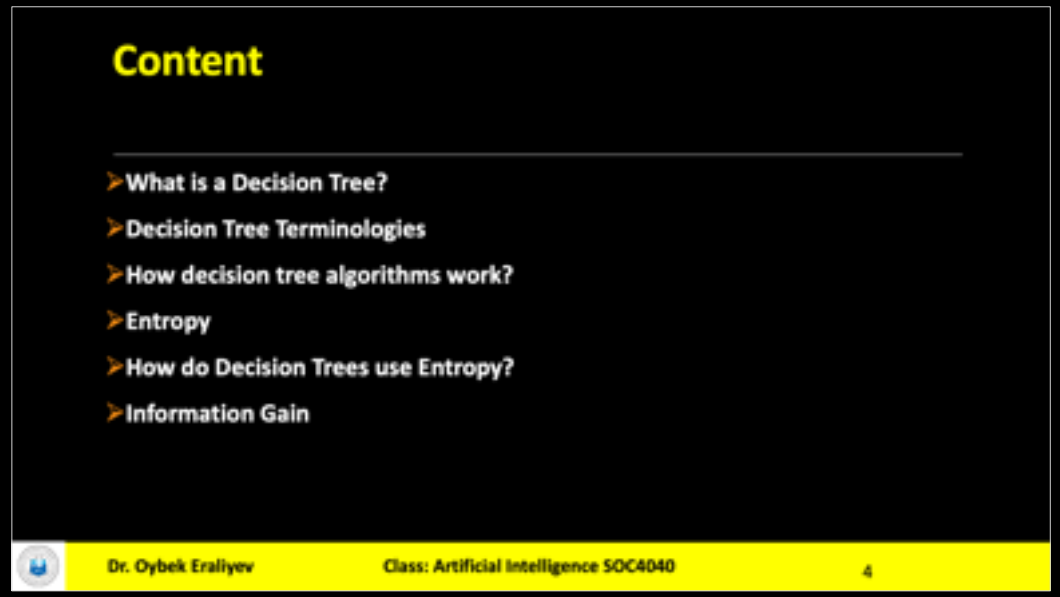
Likelihood Table		Buy		
		Yes	No	
Discount	Yes	8/10	2/5	10/15
	No	2/10	3/5	5/15
		10/15	5/15	

What is the conditional probobility of customer's not making purchase at given condition?

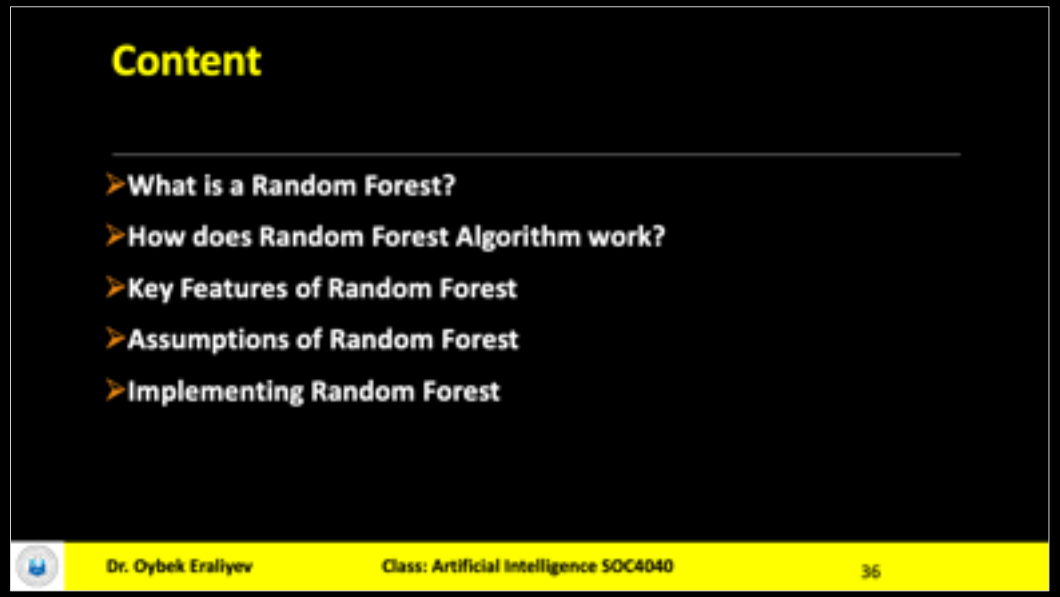
Day = Weeday
Discount = No
Free Delivery = Yes

Machine Learning Algorithms

Decision Tree



Random Forest



Content

- What is a Decision Tree?
- Decision Tree Terminologies
- How decision tree algorithms work?
- Entropy
- How do Decision Trees use Entropy?
- Information Gain

What is a Decision Tree?

Introduction

- Decision trees are a popular machine learning algorithm that can be used for both regression and classification tasks.

- They are easy to understand, interpret, and implement, making them an ideal choice for beginners in the field of machine learning.

- In this lecture, we will cover all aspects of the decision tree algorithm, including the working principles, different types of decision trees, the process of building decision trees, and how to evaluate and optimize these.

What is a Decision Tree?

Introduction

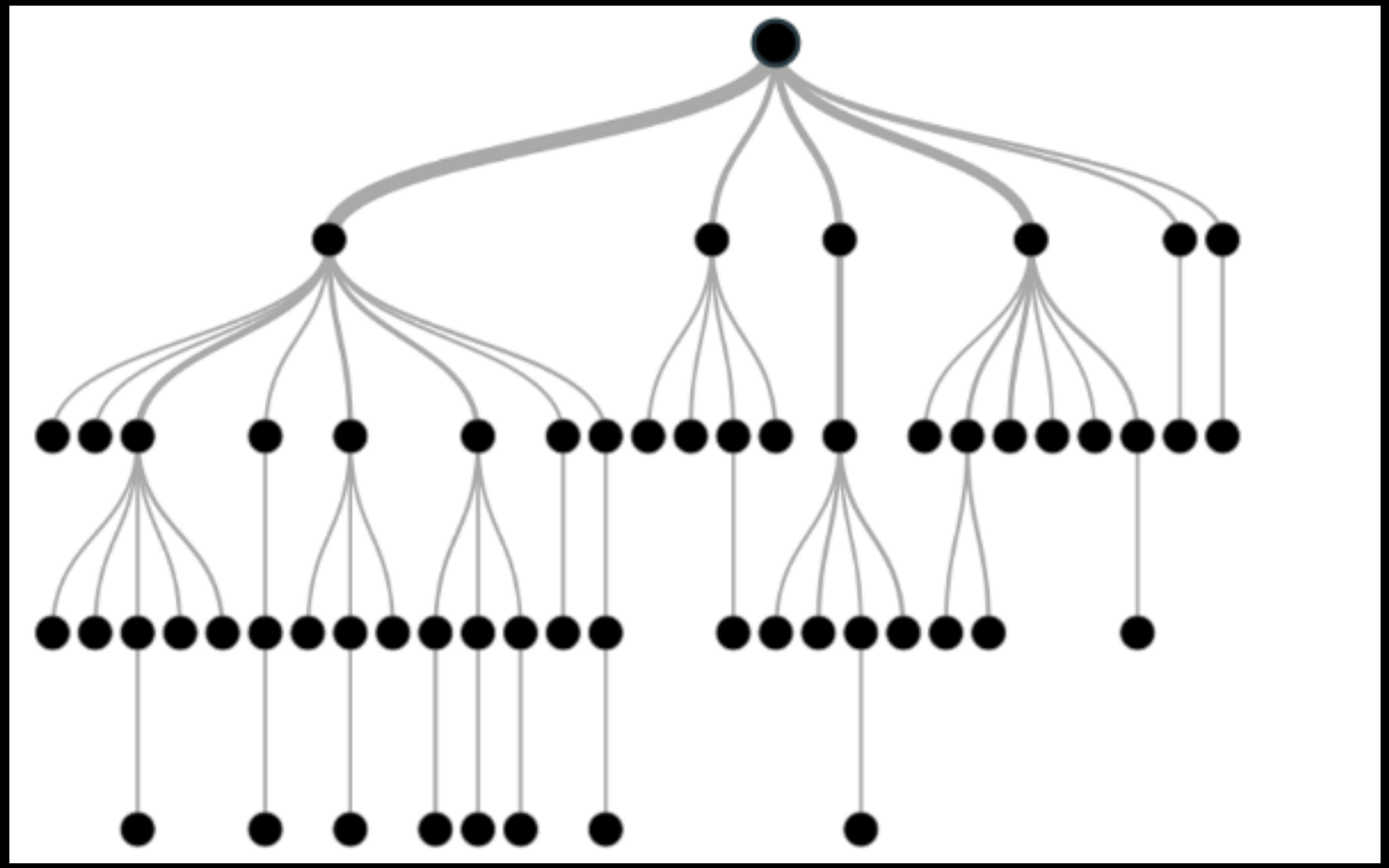
- A decision tree, which has a hierarchical structure made up of root, branches, internal, and leaf nodes, is a non-parametric supervised learning approach used for classification and regression applications.

- It is a tool that has applications spanning several different areas. The name itself suggests that it uses a flowchart like a tree structure to show the predictions that result from a series of feature-based splits.

- It starts with a root node and ends with a decision made by leaves.

What is a Decision Tree?

Introduction



What is a Decision Tree?

Types of Decision Tree

- ID3: This algorithm measures how mixed up the data is at a node using something called entropy. It then chooses the feature that helps to clarify the data the most.

- C4.5: This is an improved version of ID3 that can handle missing data and continuous attributes.

- CART: This algorithm uses a different measure called Gini impurity to decide how to split the data. It can be used for both classification (sorting data into categories) and regression (predicting continuous values) tasks.

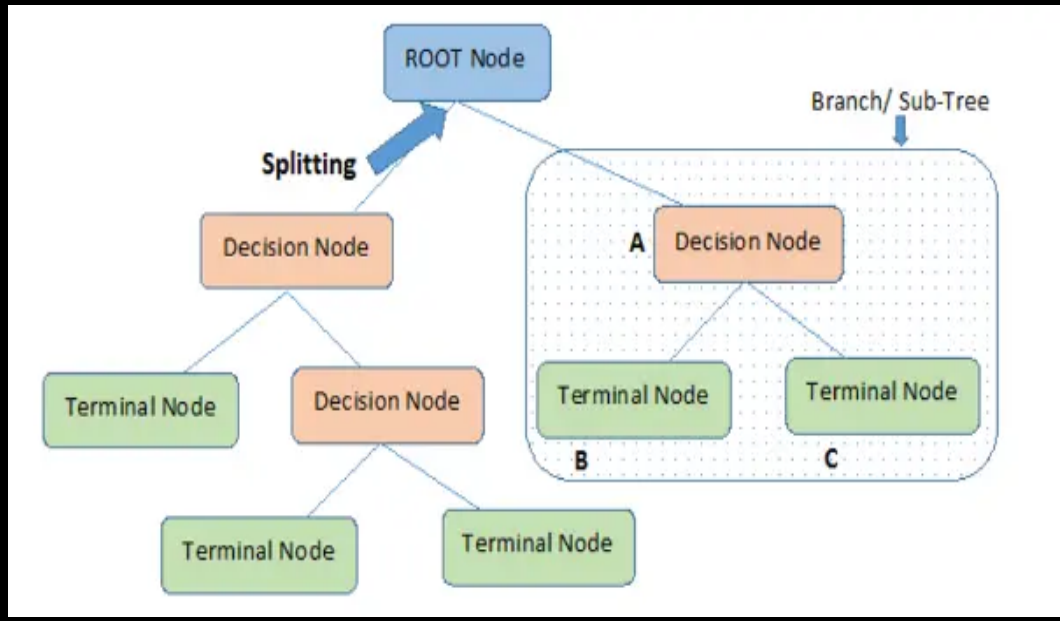
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Decision Tree Terminologies

- Root Node: The initial node at the beginning of a decision tree, where the entire population or dataset starts dividing based on various features or conditions.

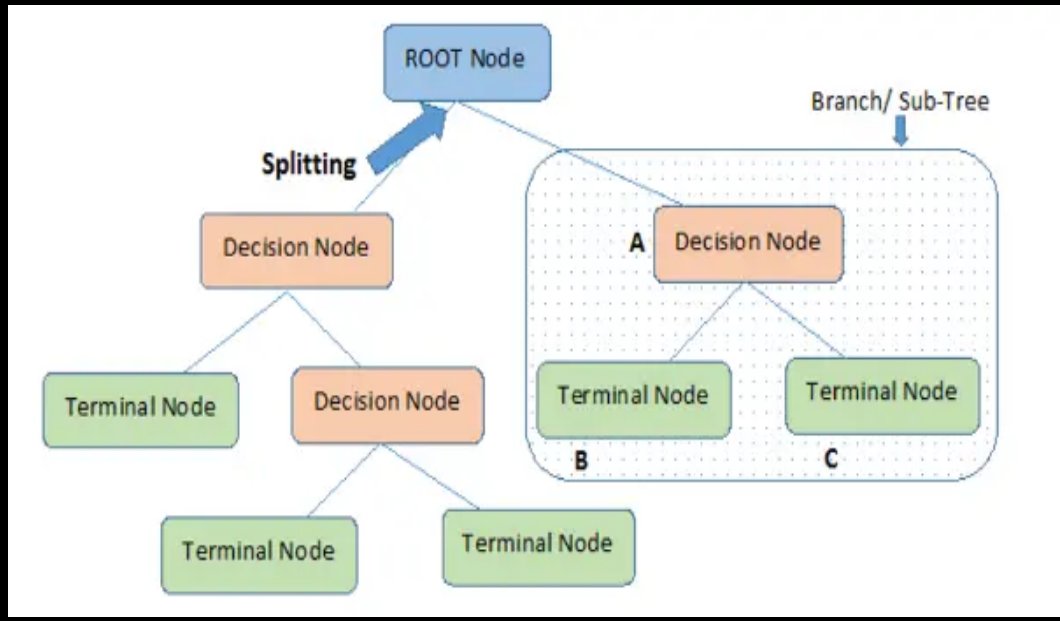
- Decision Nodes: Nodes resulting from the splitting of root nodes are known as decision nodes. These nodes represent intermediate decisions or conditions within the tree.



Decision Tree Terminologies

- Leaf Nodes: Nodes where further splitting is not possible, often indicating the final classification or outcome. Leaf nodes are also referred to as terminal nodes.

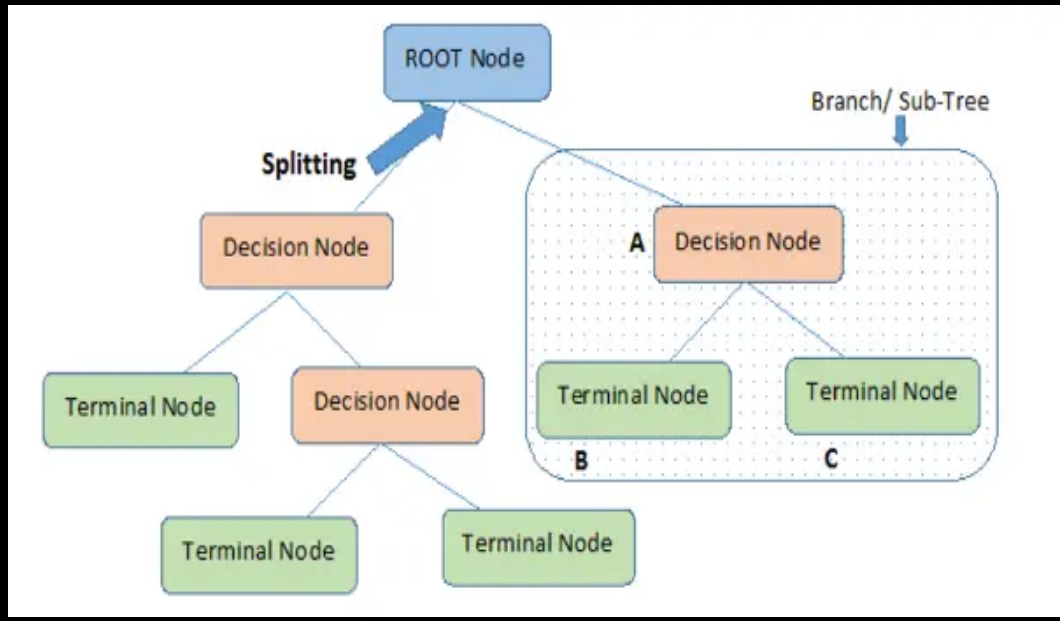
- Sub-Tree: Similar to a subsection of a graph being called a sub-graph, a sub-section of a decision tree is referred to as a sub-tree. It represents a specific portion of the decision tree.



Decision Tree Terminologies

- Pruning: The process of removing or cutting down specific nodes in a decision tree to prevent overfitting and simplify the model.

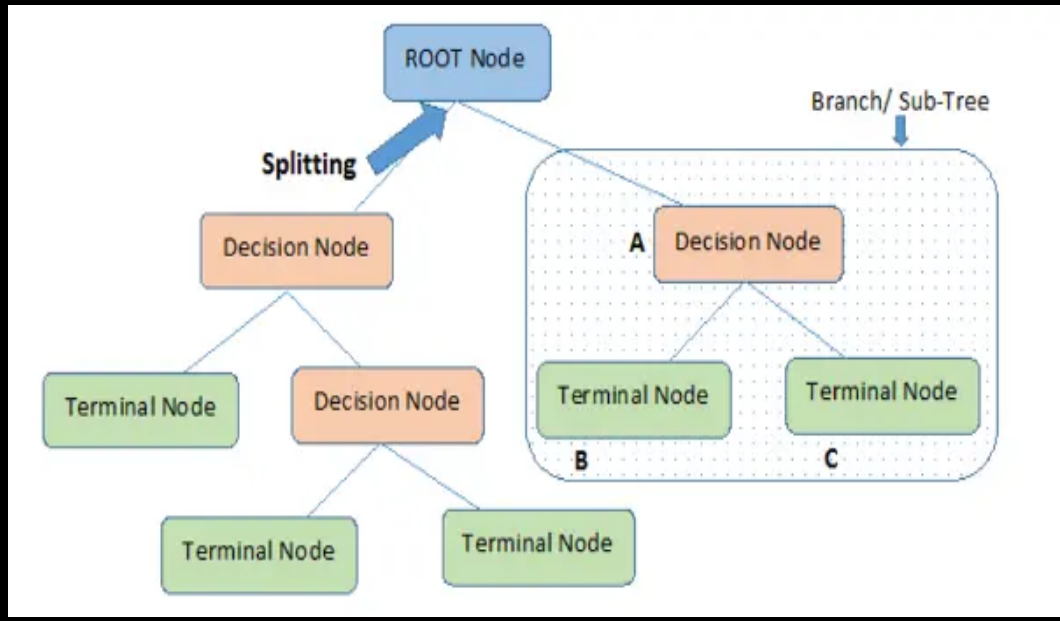
- Branch / Sub-Tree: A subsection of the entire decision tree is referred to as a branch or sub-tree. It represents a specific path of decisions and outcomes within the tree.



Decision Tree Terminologies

- Parent and Child Node: In a decision tree, a node that is divided into sub-nodes is known as a parent node, and the sub-nodes emerging from it are referred to as child nodes.

- The parent node represents a decision or condition, while the child nodes represent the potential outcomes or further decisions based on that condition.



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How decision tree algorithms work?

Decision Tree algorithm works in simpler steps:

- Starting at the Root: The algorithm begins at the top, called the "root node," representing the entire dataset.
- Asking the Best Questions: It looks for the most important feature or question that splits the data into the most distinct groups. This is like asking a question at a fork in the tree.
- Branching Out: Based on the answer to that question, it divides the data into smaller subsets, creating new branches. Each branch represents a possible route through the tree.
- Repeating the Process: The algorithm continues asking questions and splitting the data at each branch until it reaches the final "leaf nodes," representing the predicted outcomes or classifications.

How decision tree algorithms work?

Decision Tree Assumptions

➢ Several assumptions are made to build effective models when creating decision trees. These assumptions help guide the tree's construction and impact its performance. Here are some common assumptions and considerations when creating decision trees:

- Binary Splits: Decision trees typically make binary splits, meaning each node divides the data into two subsets based on a single feature or condition. This assumes that each decision can be represented as a binary choice.
- Recursive Partitioning: Decision trees use a recursive partitioning process, where each node is divided into child nodes, and this process continues until a stopping criterion is met. This assumes that data can be effectively subdivided into smaller, more manageable subsets.

How decision tree algorithms work?

Decision Tree Assumptions

- Feature Independence: Decision trees often assume that the features used for splitting nodes are independent. In practice, feature independence may not hold, but decision trees can still perform well if features are correlated.

- Homogeneity: Decision trees aim to create homogeneous subgroups in each node, meaning that the samples within a node are as similar as possible regarding the target variable. This assumption helps in achieving clear decision boundaries.

How decision tree algorithms work?

Decision Tree Assumptions

- Top-Down Greedy Approach: Decision trees are constructed using a top-down, greedy approach, where each split is chosen to maximize information gain or minimize impurity at the current node. This may not always result in the globally optimal tree.

- Categorical and Numerical Features: Decision trees can handle both categorical and numerical features. However, they may require different splitting strategies for each type.

How decision tree algorithms work?

Decision Tree Assumptions

- Overfitting: Decision trees are prone to overfitting when they capture noise in the data. Pruning and setting appropriate stopping criteria are used to address this assumption.

- Impurity Measures: Decision trees use impurity measures such as Gini impurity or entropy to evaluate how well a split separates classes. The choice of impurity measure can impact tree construction.

- No Missing Values: Decision trees assume that there are no missing values in the dataset or that missing values have been appropriately handled through imputation or other methods.

How decision tree algorithms work?

Decision Tree Assumptions

- Equal Importance of Features: Decision trees may assume equal importance for all features unless feature scaling or weighting is applied to emphasize certain features.

- No Outliers: Decision trees are sensitive to outliers, and extreme values can influence their construction. Preprocessing or robust methods may be needed to handle outliers effectively.

- Sensitivity to Sample Size: Small datasets may lead to overfitting, and large datasets may result in overly complex trees. The sample size and tree depth should be balanced.

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- Information Gain
- Pruning

Entropy

➢ The formula for Entropy is shown below:

$$E(S) = -p_{(+)}logp_{(+)} - p_{(-)}logp_{(-)}$$

➢ Here,

- $p_{(+)}$ is the probability of positive class
- $p_{(-)}$ is the probability of negative class
- S is the subset of the training example

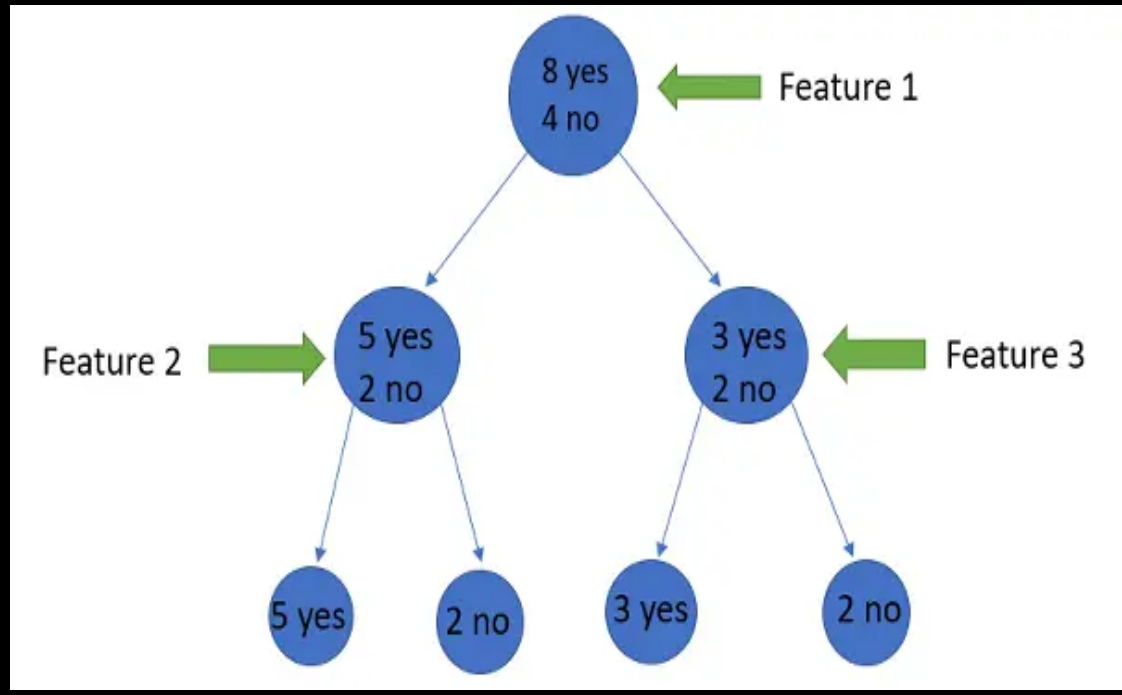
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How do Decision Trees use Entropy?

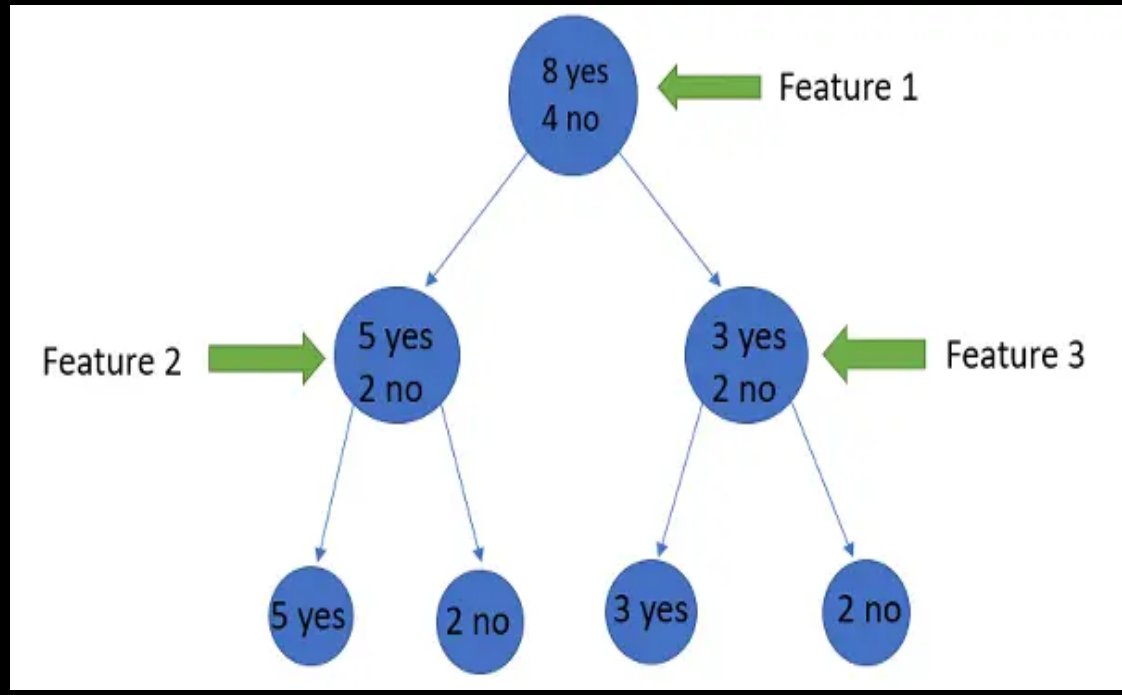
- Entropy basically measures the impurity of a node. Impurity is the degree of randomness; it tells how random our data is.

- A pure sub-split means that either you should be getting "yes", or you should be getting "no".



How do Decision Trees use Entropy?

- Suppose a feature has 8 “yes” and 4 “no” initially, after the first split the left node gets 5 ‘yes’ and 2 ‘no’ whereas right node gets 3 ‘yes’ and 2 ‘no’.
- We see here the split is not pure, why? Because we can still see some negative classes in both the nodes. In order to make a decision tree, we need to calculate the impurity of each split, and when the purity is 100%, we make it as a leaf node.



How do Decision Trees use Entropy?

➢ To check the impurity of feature 2 and feature 3 we will take the help for Entropy formula.

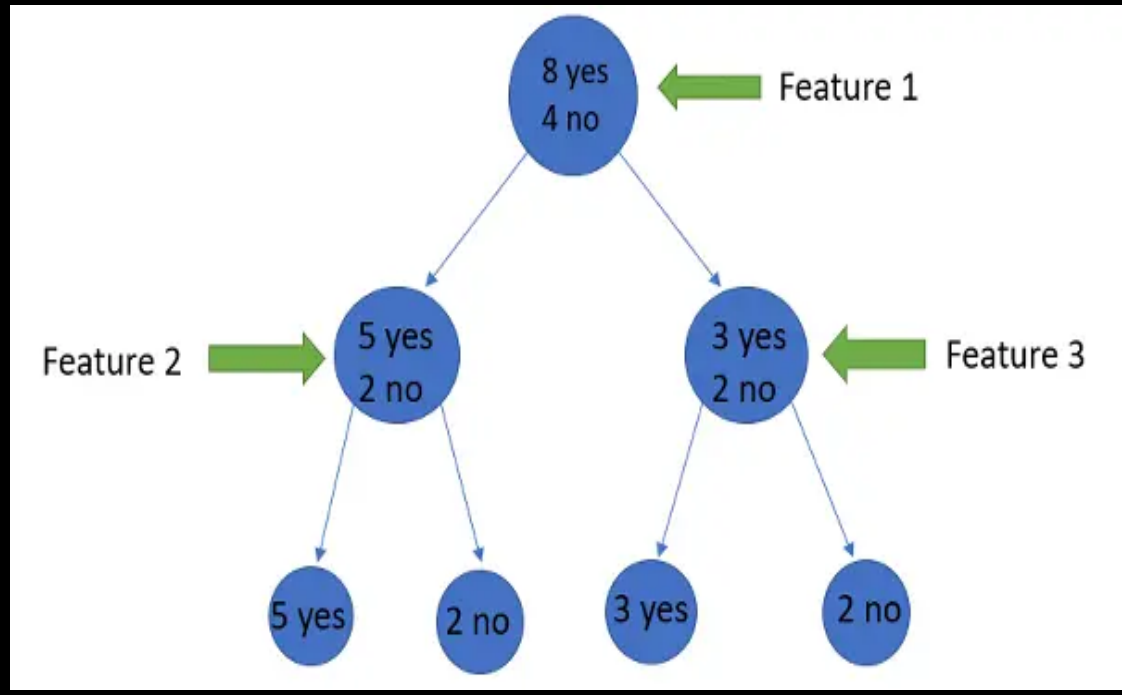
$$\Rightarrow \quad -\left(\frac{5}{7}\right)log_2\left(\frac{5}{7}\right) - \left(\frac{2}{7}\right)log_2\left(\frac{2}{7}\right)$$

$$\Rightarrow \quad -(0.71 * -0.49) - (0.28 * -1.83)$$

$$\Rightarrow \quad -(-0.34) - (-0.51)$$

$$\Rightarrow \quad 0.34 + 0.51$$

$$\Rightarrow \quad 0.85$$



How do Decision Trees use Entropy?

➢ To check the impurity of feature 2 and feature 3 we will take the help for Entropy formula.

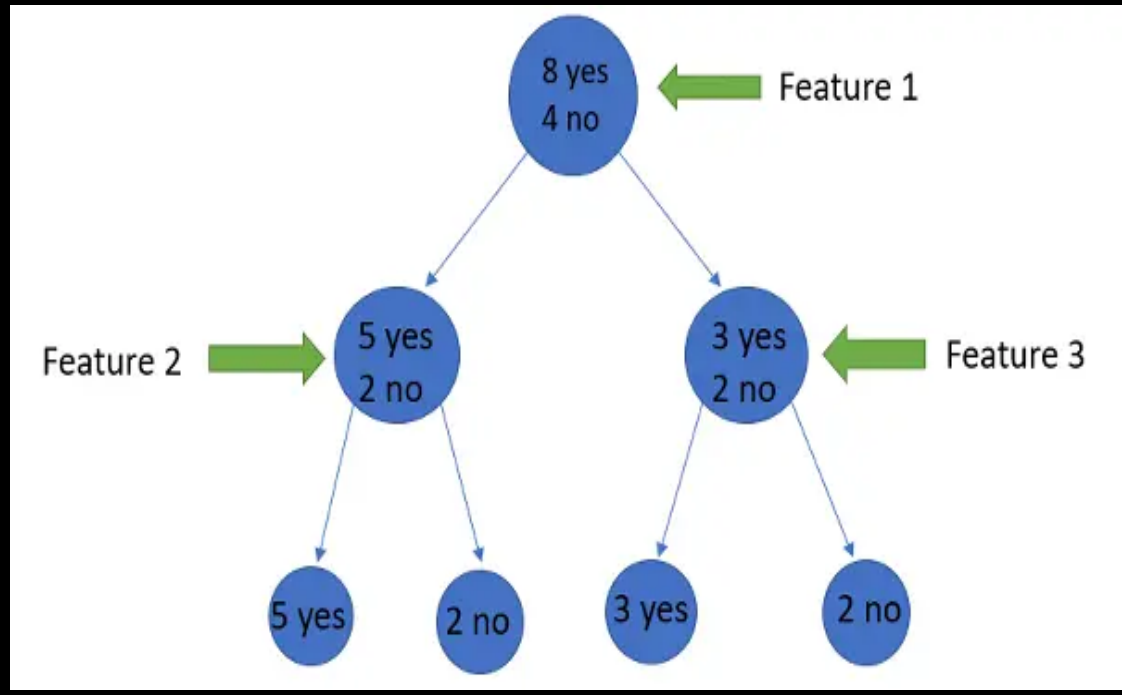
$$\Rightarrow \quad -\left(\frac{3}{5}\right)log_2\left(\frac{3}{5}\right) - \left(\frac{2}{5}\right)log_2\left(\frac{2}{5}\right)$$

$$\Rightarrow \quad -(0.6 * -0.73) - (0.4 * -1.32)$$

$$\Rightarrow \quad -(-0.438) - (-0.528)$$

$$\Rightarrow \quad 0.438 + 0.528$$

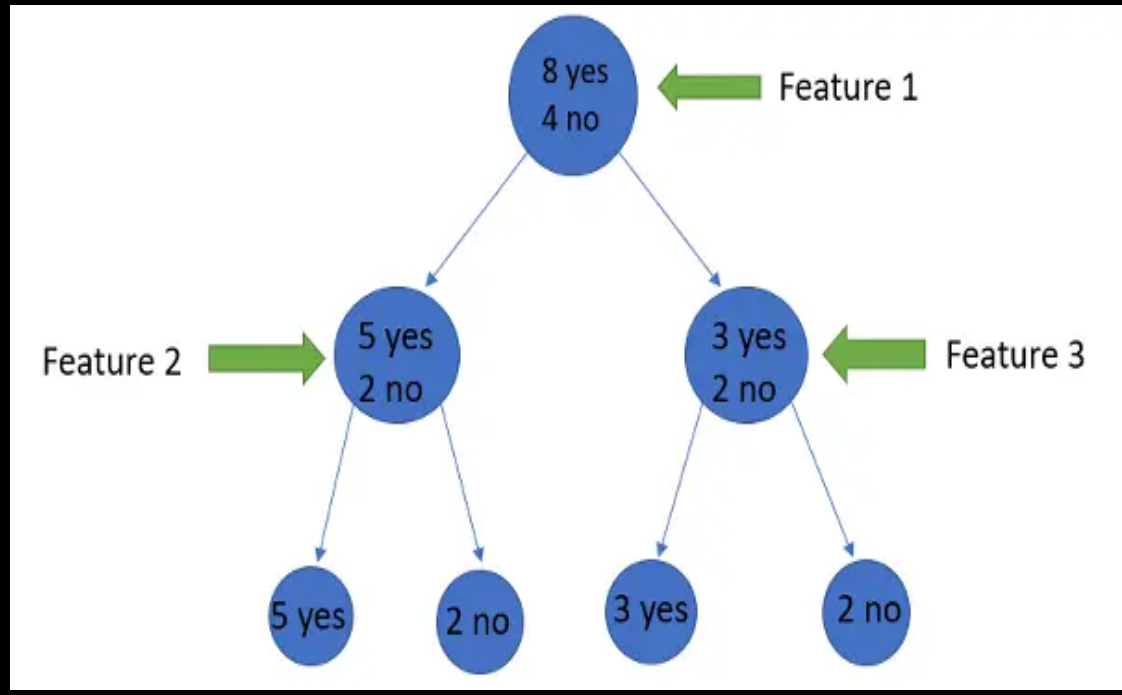
$$\Rightarrow \quad 0.966$$



How do Decision Trees use Entropy?

- We can clearly see from the tree itself that left node has low entropy or more purity than right node since left node has a greater number of "yes" and it is easy to decide here.

- Always remember that the higher the Entropy, the lower will be the purity and the higher will be the impurity.



How do Decision Trees use Entropy?

- As mentioned earlier the goal of machine learning is to decrease the uncertainty or impurity in the dataset, here by using the entropy we are getting the impurity of a particular node, we don't know if the parent entropy or the entropy of a particular node has decreased or not.

- For this, we bring a new metric called "Information gain" which tells us how much the parent entropy has decreased after splitting it with some feature.

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Information Gain

➢ Information gain measures the reduction of uncertainty given some feature and it is also a deciding factor for which attribute should be selected as a decision node or root node.

$$Information\ Gain = E(Y) - E(Y|X)$$

$$I.G(S,a) = Entropy(S) - \sum_{v \in values(a)} \frac{|S_v|}{|S|} Entropy(S_v)$$

➢ It is just entropy of the full dataset – entropy of the dataset given some feature.

Information Gain

Dataset for implementation of Decision Tree Model

A	B	C	D	E	F
Day	**Outlook**	**Temp**	**Humidity**	**Wind**	**PlayTennis**
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Information Gain

Dataset for implementation of Decision Tree Model

H	I	J	K	L	M
Day	**Outlook**	**Temp**	**Humidity**	**Wind**	**PlayTennis**
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes

For Left Node

O	P	Q	R	S	T
Day	**Outlook**	**Temp**	**Humidity**	**Wind**	**PlayTennis**
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D10	Rain	Mild	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

For Right Node

Information Gain

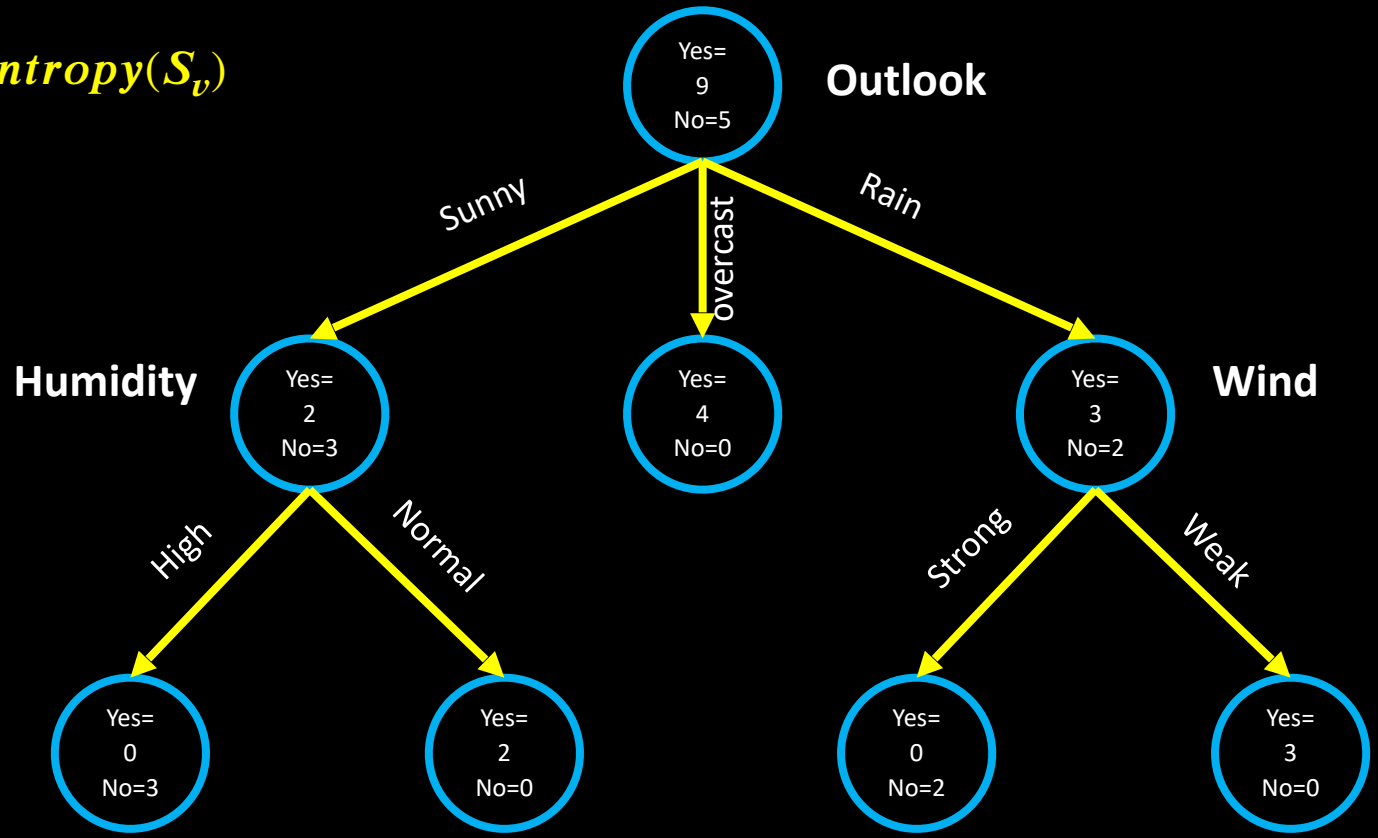
Dataset for implementation of Decision Tree Model

$$I.G(S,a) = Entropy(S) - \sum_{v \in values(a)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$E(S) = -p_{(+)} log p_{(+)} - p_{(-)} log p_{(-)}$$

X_test = {Outlook: sunny, Hum: High,
Tep: Hot, Wind: weak}

what is the prediction?



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- What is a Random Forest?
- How does Random Forest Algorithm work?
- Key Features of Random Forest
- Assumptions of Random Forest
- Implementing Random Forest

What is a Random Forest?

- Random Forest is a machine learning algorithm that uses many decision trees to make better predictions. Each tree looks at different random parts of the data and their results are combined by voting for classification or averaging for regression which makes it as ensemble learning technique.
- This helps in improving accuracy and reducing errors.

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How does Random Forest Algorithm work?

- **Create Many Decision Trees:** The algorithm makes many decision trees each using a random part of the data. So every tree is a bit different.
- **Pick Random Features:** When building each tree it doesn't look at all the features (columns) at once. It picks a few at random to decide how to split the data. This helps the trees stay different from each other.
- **Each Tree Makes a Prediction:** Every tree gives its own answer or prediction based on what it learned from its part of the data.

How does Random Forest Algorithm work?

- **Combine the Predictions:** For **classification** we choose a category as the final answer is the one that most trees agree on i.e majority voting and for **regression** we predict a number as the final answer is the average of all the trees predictions.
- **Why It Works Well:** Using random data and features for each tree helps avoid overfitting and makes the overall prediction more accurate and trustworthy.

How does Random Forest Algorithm work?

Original dataset

A	B	C	D	E	F
Day	**Outlook**	**Temp**	**Humidity**	**Wind**	**PlayTennis**
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

How does Random Forest Algorithm work?

Bootstrapped dataset 1

A	B	C	D	E	F
Day	**Outlook**		**Humidity**	**Wind**	**PlayTennis**
D1	Sunny		High	Weak	No
D2	Sunny		High	Strong	No
D3	Overcast		High	Weak	Yes
D5	Rain		Normal	Weak	Yes
D5	Rain		Normal	Weak	Yes
D6	Rain		Normal	Strong	No
D7	Overcast		Normal	Strong	Yes
D8	Sunny		High	Weak	No
D9	Sunny		Normal	Weak	Yes
D10	Rain		Normal	Weak	Yes
D7	Overcast		Normal	Strong	Yes
D12	Overcast		High	Strong	Yes
D13	Overcast		Normal	Weak	Yes
D14	Rain		High	Strong	No

Bootstrapped dataset 2

A	B	C	D	E	F
Day	**Outlook**	**Temp**	**Humidity**	**Wind**	**PlayTennis**
D3	Overcast	Hot	High	Weak	Yes
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D6	Rain	Cool	Normal	Strong	No
D14	Rain	Mild	High	Strong	No

Bootstrapped dataset 3

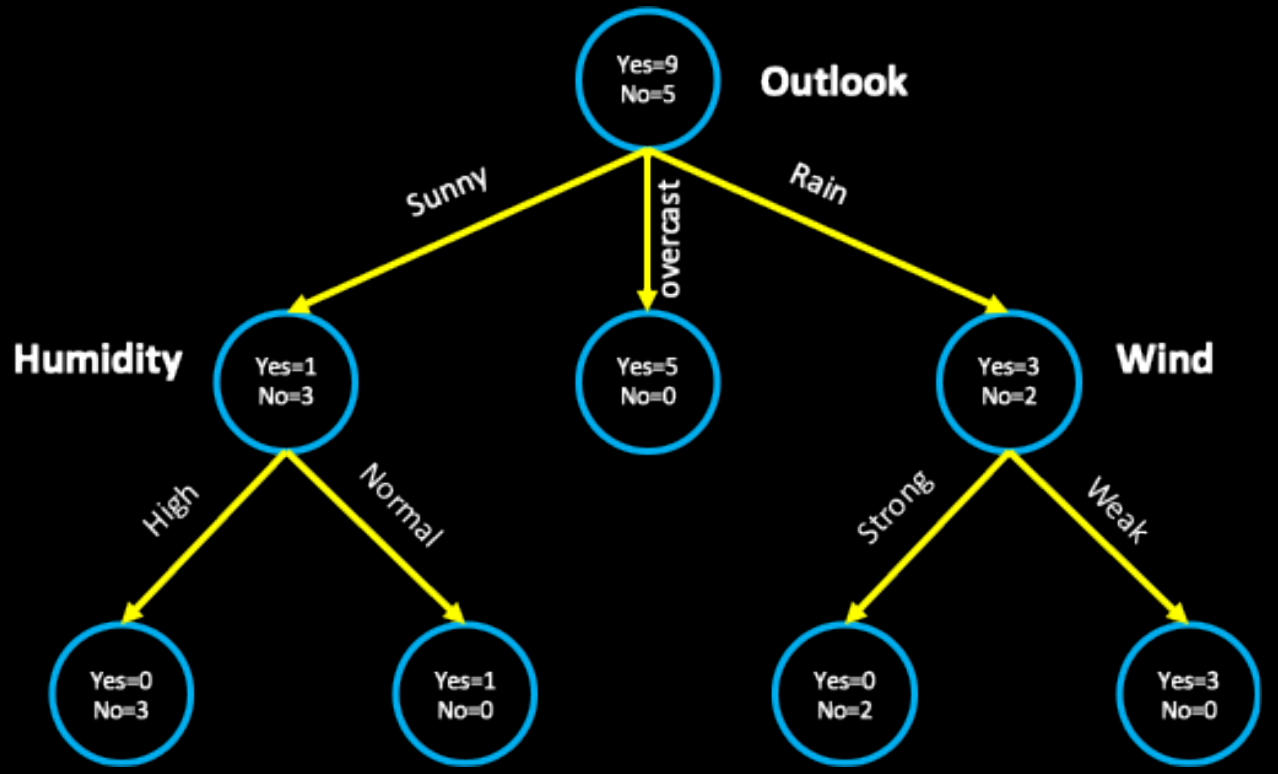
A	B	C	D	E	F
Day	**Outlook**	**Temp**	**Humidity**		**PlayTennis**
D1	Sunny	Hot	High		No
D2	Sunny	Hot	High		No
D1	Sunny	Hot	High		No
D4	Rain	Mild	High		Yes
D5	Rain	Cool	Normal		Yes
D6	Rain	Cool	Normal		No
D7	Overcast	Cool	Normal		Yes
D9	Sunny	Cool	Normal		Yes
D9	Sunny	Cool	Normal		Yes
D10	Rain	Mild	Normal		Yes
D11	Sunny	Mild	Normal		Yes
D12	Overcast	Mild	High		Yes
D13	Overcast	Hot	Normal		Yes
D14	Rain	Mild	High		No

How does Random Forest Algorithm work?

Bootstrapped dataset 1

A	B	C	D	E	F
Day	**Outlook**		**Humidity**	**Wind**	**PlayTennis**
D1	Sunny		High	Weak	No
D2	Sunny		High	Strong	No
D3	Overcast		High	Weak	Yes
D5	Rain		Normal	Weak	Yes
D5	Rain		Normal	Weak	Yes
D6	Rain		Normal	Strong	No
D7	Overcast		Normal	Strong	Yes
D8	Sunny		High	Weak	No
D9	Sunny		Normal	Weak	Yes
D10	Rain		Normal	Weak	Yes
D7	Overcast		Normal	Strong	Yes
D12	Overcast		High	Strong	Yes
D13	Overcast		Normal	Weak	Yes
D14	Rain		High	Strong	No

Decision Tree 1

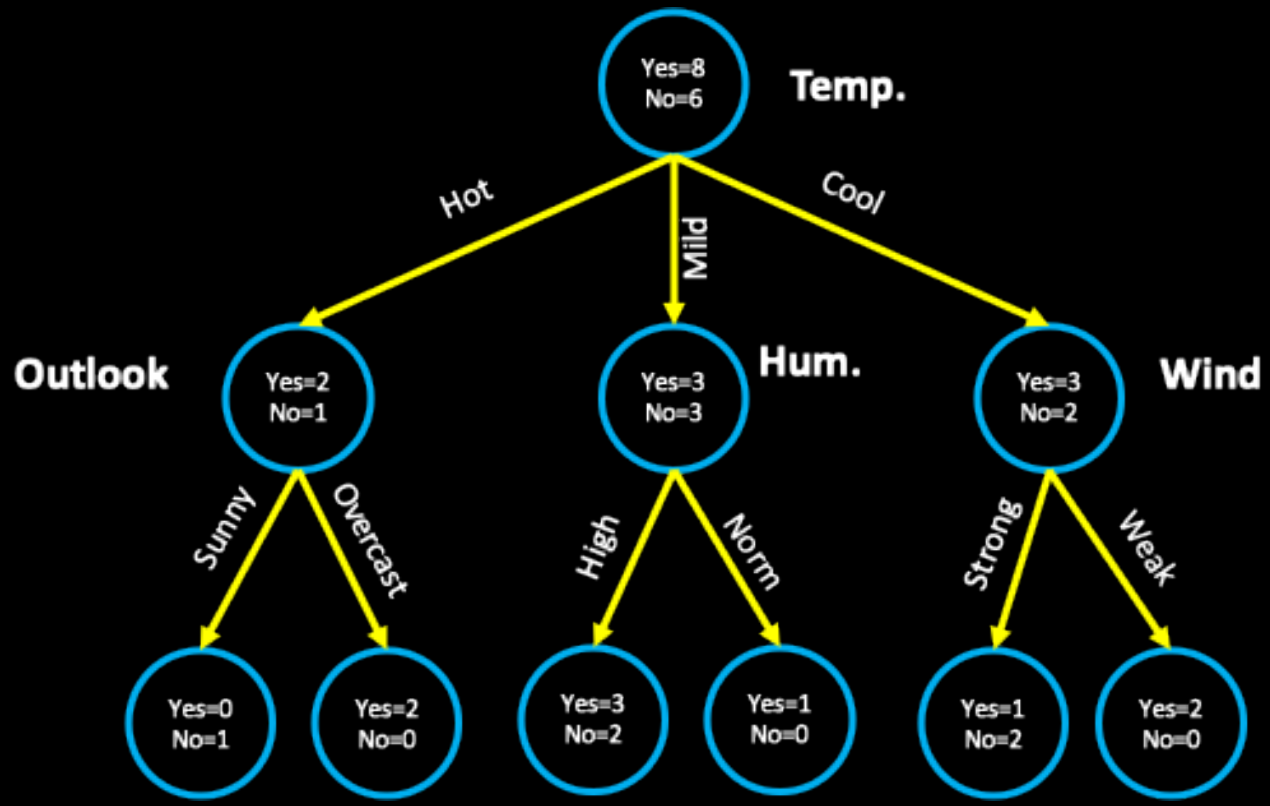


How does Random Forest Algorithm work?

Bootstrapped dataset 2

A	B	C	D	E	F
Day	**Outlook**	**Temp**	**Humidity**	**Wind**	**PlayTennis**
D3	Overcast	Hot	High	Weak	Yes
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D6	Rain	Cool	Normal	Strong	No
D14	Rain	Mild	High	Strong	No

Decision Tree 2

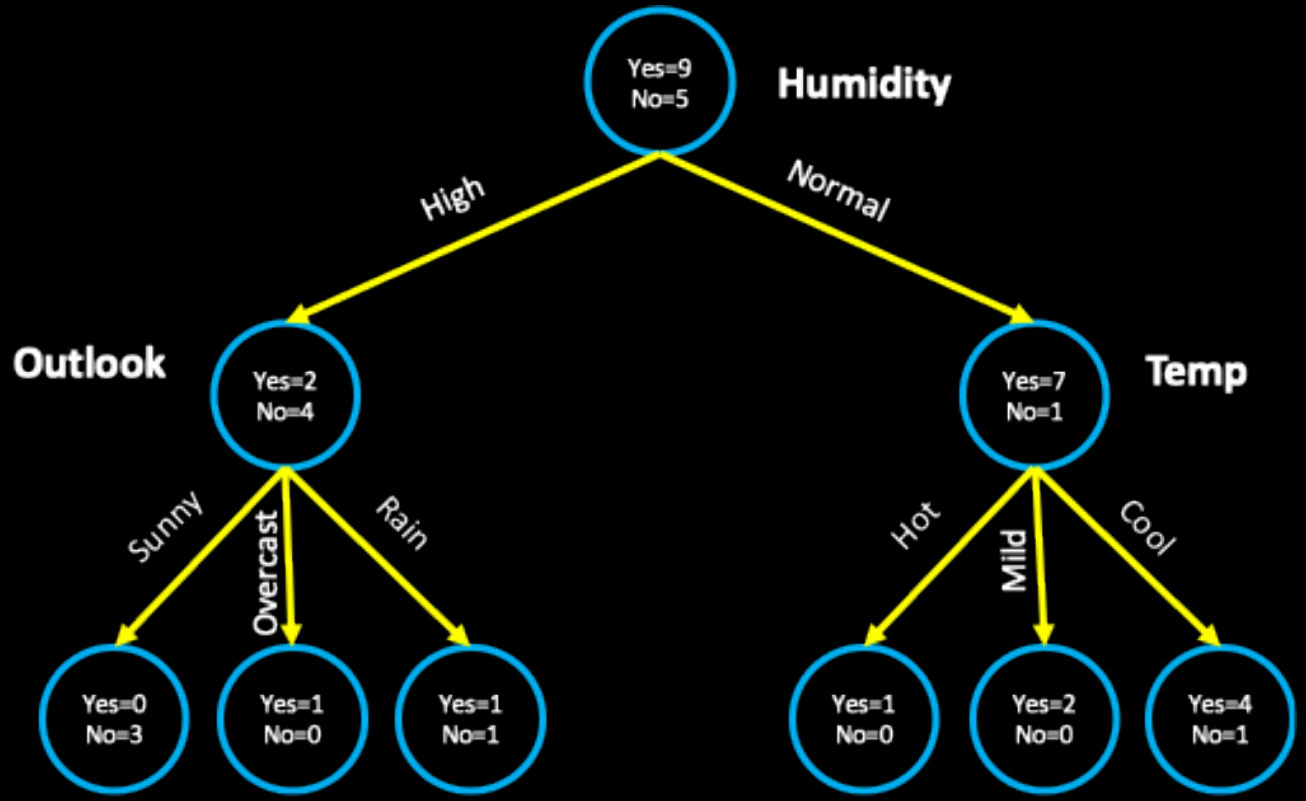


How does Random Forest Algorithm work?

Bootstrapped dataset 3

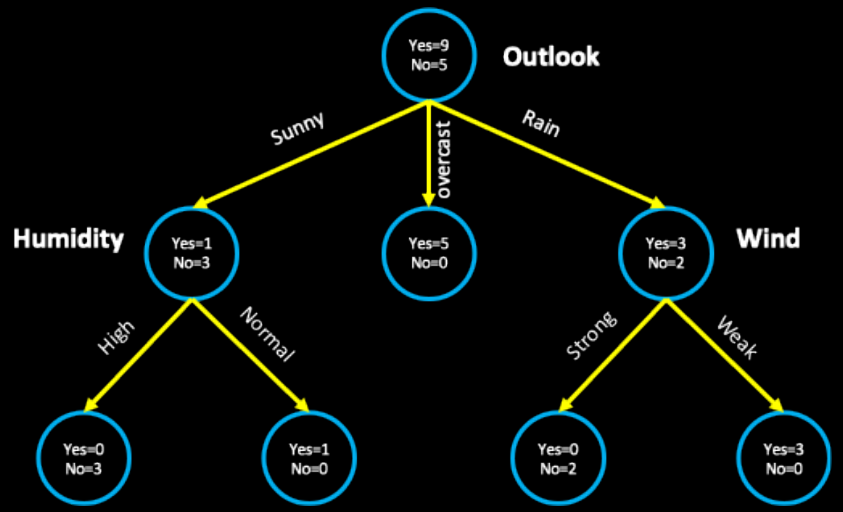
A	B	C	D	E	F
Day	**Outlook**	**Temp**	**Humidity**		**PlayTennis**
D1	Sunny	Hot	High		No
D2	Sunny	Hot	High		No
D1	Sunny	Hot	High		No
D4	Rain	Mild	High		Yes
D5	Rain	Cool	Normal		Yes
D6	Rain	Cool	Normal		No
D7	Overcast	Cool	Normal		Yes
D9	Sunny	Cool	Normal		Yes
D9	Sunny	Cool	Normal		Yes
D10	Rain	Mild	Normal		Yes
D11	Sunny	Mild	Normal		Yes
D12	Overcast	Mild	High		Yes
D13	Overcast	Hot	Normal		Yes
D14	Rain	Mild	High		No

Decision Tree 3

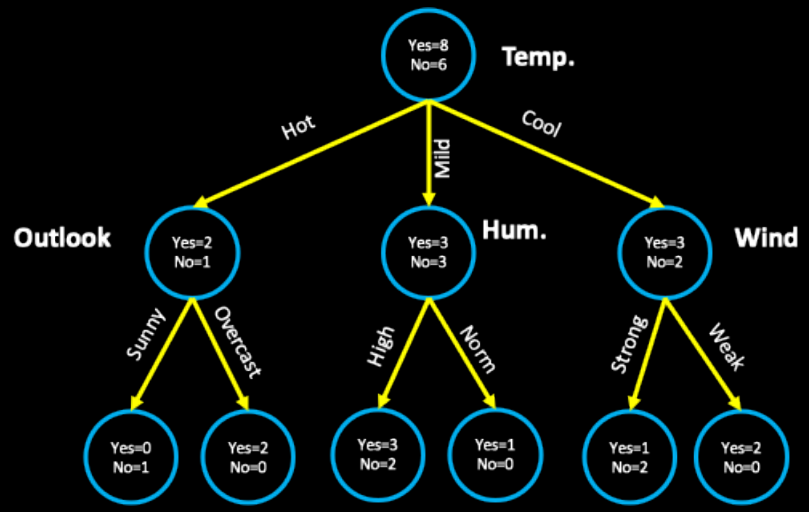


How does Random Forest Algorithm work?

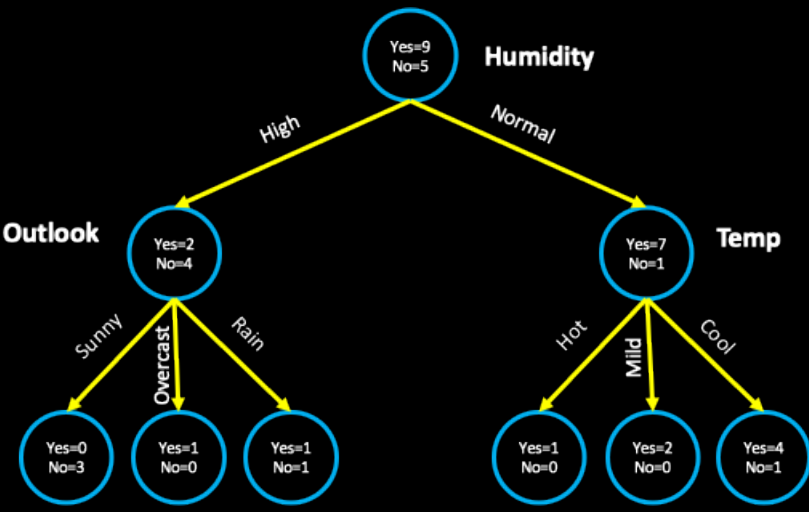
Decision Tree 1



Decision Tree 2



Decision Tree 3



Testting: x = {Outlook=Sunny, Temp=Cool, Humidity=High, Wind=Strong}

DT 1: No, DT 2: No, DT 3: No

Final prediction of Random Forest: No

Content

- ~~What is a Random Forest?~~
- ~~How does Random Forest Algorithm work?~~
- Key Features of Random Forest
- Assumptions of Random Forest
- Implementing Random Forest

Key Features of Random Forest

- **Handles Missing Data:** It can work even if some data is missing so you don’t always need to fill in the gaps yourself.
- **Shows Feature Importance:** It tells you which features (columns) are most useful for making predictions which helps you understand your data better.
- **Works Well with Big and Complex Data:** It can handle large datasets with many features without slowing down or losing accuracy.
- **Used for Different Tasks:** You can use it for both **classification** like predicting types or labels and **regression** like predicting numbers or amounts.

Content

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Assumptions of Random Forest

- **Each tree makes its own decisions**: Every tree in the forest makes its own predictions without relying on others.
- **Random parts of the data are used**: Each tree is built using random samples and features to reduce mistakes.
- **Enough data is needed**: Sufficient data ensures the trees are different and learn unique patterns and variety.
- **Different predictions improve accuracy**: Combining the predictions from different trees leads to a more accurate final result.

Content

- ~~What is a Random Forest?~~
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THANK YOU